Coherence of Resilience Measures for the Atlantic Meridional Overturning Circulation Across a Hierarchy of Models

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Abstract

The Atlantic Meridional Overturning Circulation (AMOC) is a critical component of the global climate system, and its potential collapse one of the most consequential climate tipping events. Quantifying AMOC resilience, that is, its capacity to maintain the active circulation state under external perturbations, is therefore of high scientific and societal importance. Resilience is a multifaceted concept, encompassing many notions of transient system response. Whether these distinct measures yield consistent, and therefore trustworthy, signals across models of different complexity has not been systematically examined. Here, we address this question by analysing four models that span a hierarchy in physical complexity: (i) a 3-box salinity model, (ii) a two-dimensional latitude-depth Boussinesq model of the overturning circulation, (iii) the Earth system model of intermediate complexity CLIMBER-X, and (iv) the general circulation model PlaSim-LSG. Across the hierarchy, we compute model-specific implementations of four complementary resilience concepts: local resilience, convergence time, basin volume, and minimal critical shock. Our results suggest that not all measures are coherent across the

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model hierarchy, as exhibited by incompatible trends under increasing CO_2 forcing in each model. This points to deficiencies in model compatibility and challenges in applying these concepts in real-world systems. Several measures indicate an unexpected increase in system resilience under climate change. These findings have direct implications for the detectability of an imminent AMOC collapse from observational or model-based early-warning signals.

Keywords: Atlantic Meridional Overturning Circulation, tipping points, resilience, model hierarchy

1 Introduction

The Atlantic Meridional Overturning Circulation is one of the dominant features of the Earth's oceans, transporting large amounts of warm, saline surface waters northward across the Atlantic and returning cold, deep North Atlantic water southward at depth ~~CCC~~(Talley et al., 2003; Johns et al., 2023). This large-scale transport pattern underpins the relatively mild climate of Western Europe and plays a central role in the global carbon and heat cycles ~~CCC~~(Ganachaud and Wunsch, 2000; Jackson et al., 2015; Schaumann and Alastrué de Asenjo, 2025). Palaeoclimate records document repeated rapid transitions between an active (AMOC-on) and a weakened or collapsed (AMOC-off) state during the Pleistocene, associated with Dansgaard–Oeschger events ~~CCC~~(Dansgaard et al., 1984; Clark et al., 2002; Menviel et al., 2020). Under continued anthropogenic forcing, rising sea surface temperatures, freshening of the North Atlantic from increased precipitation, and Greenland ice-sheet melt may push the AMOC towards a similar transition ~~CCC~~(Yang et al., 2016; Lohmann and Ditlevsen, 2021; Armstrong McKay et al., 2022; van Westen et al., 2024).

More concretely, in background climate configurations where at least two stable equilibria coexist (the on and off states), the system can be permanently tipped from one to the other by crossing a critical threshold leading to the annihilation of one of the formerly stable states (bifurcation-induced tipping) or by a sufficiently large state perturbation (shock-induced tipping) (Ashwin et al., 2012; Alkhayuon et al., 2019). Rate-induced tipping is another pathway, though it will not be relevant for this work. In the bifurcation-theory framework for AMOC tipping, the transition to the off state is often conceptualised as a saddle-node bifurcation (van Westen et al., 2025). A shock-induced transition, on the other hand, can be understood as an exit of the system state from the basin of attraction associated with the AMOC on state. In both frameworks, the AMOC is investigated as an autonomous dynamical system that may be influenced by slow parameter changes or external perturbations, such as noise. This allows for the characterisation of the system in terms of several quantities, subsumed under the term "resilience" (Krakovská et al., 2024).

One important notion of resilience has been applied to the AMOC in the context of a bifurcation-induced transition. As the system approaches such a transition through a gradual change in the background climate, the stable on-state loses linear stability, a phenomenon known as critical slowing down (CSD) (Scheffer et al., 2009). CSD manifests in observable statistics: near the bifurcation, the dominant eigenvalue of the linearised system approaches zero, leading to increases in variance and autocorrelation of system fluctuations (Held and Kleinen, 2004; Dakos et al., 2008). These generic signatures have motivated a large body of research on early-warning indicators of tipping (Lenton, 2011; Boers, 2021; Ditlevsen and Ditlevsen, 2023).

Yet CSD-based indicators capture only the system's reaction to small perturbations around the equilibrium, known as local resilience, quantifying how the on-state changes locally as the bifurcation is approached. A complementary perspective is provided by non-local resilience concepts, which characterise the size and geometry of the basin of attraction of the on-state and therefore also capture the system's capacity to recover from larger perturbations (Holling, 1973; Gunderson, 2000; Menck et al., 2013). A comprehensive assessment of the resilience of the AMOC on state should encompass many different notions. In this work, we consider four resilience measures with the following heuristic definitions and interpretations. The concrete implementation and calculations for each model will vary and are explained in Section 2.

1. **Local resilience**: capacity of a system to continuously absorb small perturbations to its equilibium on-state, such as noise and internal variability (Arnoldi et al., 2016).

2. **Convergence time**: duration trajectories take to return to the on-state equilibrium after a large perturbation (Neubert and Caswell, 1997).

3. **Basin volume**: the volume fraction of a reference region of initial model conditions that converges to the on-state (Menck et al., 2013).

4. **Minimal critical shock**: the smallest instantaneous perturbation that moves the system out of the basin of attraction of the on-state (Klinshov et al., 2015; Halekotte

and Feudel, 2020).

In natural systems, controlled perturbation experiments are rarely feasible, making direct estimation of non-local resilience quantities from observations difficult. This is especially true for the AMOC, and numerical models, therefore, play an essential role. However, climate models span an enormous range of complexity, from analytically tractable conceptual models to full Earth system models (ESMs). It is not obvious whether resilience measures computed in simple models translate to their more complex counterparts, which include many additional feedbacks and degrees of freedom (Stouffer et al., 2006; Schneider et al., 2017). If measures disagree across the hierarchy, it could mean that parts of the hierarchy do not expose them well. The extrapolation of findings from simple models to more complex ones, or to the real-world system, is then harder to justify. Several studies have examined individual resilience concepts in the context of the AMOC at specific levels of model complexity (Rahmstorf et al., 2005; Cimatoribus et al., 2014; Boers, 2021; Ditlevsen and Ditlevsen, 2023; Jacques-Dumas et al., 2024), but a systematic, multi-measure comparison across a model hierarchy is lacking.

Our analysis combines four modelling studies that span many orders of magnitude of model complexity: (i) a conceptual 3-box salinity model (Wood et al., 2019; Alkhayuon et al., 2019), (ii) a latitude-depth Boussinesq partial differential equation model (Dijkstra and Molemaker, 1997; Esclapez et al., 2026) [doi:10.1017/jfm.2025.248], (iii) the Earth system model of intermediate complexity CLIMBER-X (Willeit et al., 2022), and (iv) the coupled general circulation model PlaSim-LSG (Börner et al., 2026). Section 2 describes each model and the concrete resilience measures computed for it. Section 3 reports the individual model's results and Section 4 synthesises them across the hierarchy. In Section 4, we discuss implications for research on the AMOC and on early-warning detection of potential tipping.

2 Methods

We compute a common set of resilience measures for each model while respecting the different experimental possibilities at different levels of the hierarchy. More concretely, we analyse trajectories from perturbed initial conditions. The reduced state space in which perturbations and convergence are investigated is defined by North Atlantic and Tropical Atlantic salinity values. Salt is conserved by applying a compensating perturbation in the remaining parts of the world ocean. Figure 1 illustrates the model-specific perturbation and diagnostic regions, and their implementation is explained in the respective subsections below. Table 1 gives an overview of the concrete meaning and estimation routine for each resilience measure across the models. Initial conditions in the 3-box, Boussinesq, and CLIMBER-X experiments are probed within rectangles bounded by ± 2 psu perturbations relative to their model-specific reference configurations. This is done by drawing a large number of uniform samples from this region (3-box model) or by evenly sampling a grid of initial conditions (Boussinesq model and CLIMBER-X). PlaSim-LSG is an exception here, since we analyse existing edge-track and equilibrium runs from Börner et al. (2026) using two salinity readouts in the North and Tropical Atlantic.

We repeat the resilience analysis across a global-warming forcing coordinate, as specified for each model below. This allows us to compare directional trends of resilience, which are more meaningful across models than absolute values expressed in model-specific units. The resilience of the AMOC on-state is the primary focus throughout, though all analyses could also be carried out for other equilibria when present.

2.1 3-Box salinity model

We adopt the 5-box salinity model introduced by Wood et al. (2019) and further reduced to 3 boxes by Alkhayuon et al. (2019). The original model partitions the Atlantic–Indo-Pacific basin into five boxes: North Atlantic (N), Tropical Atlantic (T), Southern Ocean (S), Indo-Pacific (IP), and a Deep box (B). In the implemented 3-box reduction, the North and Tropical Atlantic salinity anomalies, S_N and S_T, are the two dynamical state variables. The slow variables of Southern Ocean and Deep-box salinities are prescribed

Table 1: **Overview of resilience-measure implementations across the model hierarchy.**

	3-Box model	**Boussinesq**	**CLIMBER-X**	**PlaSim-LSG**
AMOC strength	~~XXX~~ Overturning flux q (Eq. (4)) at the on-state fixed point.	~~XXX~~ Streamfunction mean over a fixed northern interior window of the $\beta_s = 0.1$ spin-up equilibrium, scaled to 18 Sv at 280 ppm.	~~XXX~~ Maximum of the Atlantic overturning streamfunction at 26°N; trailing 500-year mean of the equilibrium run.	~~XXX~~ Maximum of the Atlantic overturning streamfunction between 46° and 66°N; mean over the on-state equilibrium run.
Local resilience	$-\mathrm{Re}(\lambda_{\max})$: negative real part of the least stable eigenvalue of the Jacobian at the on-state fixed point, evaluated via forward-mode automatic differentiation (Khan and Barton, 2015).	$-\mathrm{Re}(\lambda_{\max})$: negative real part of the least stable eigenvalue fitted from a 2×2 discrete linear map via isotropic salinity perturbations.	$-\mathrm{Re}(\lambda_{\max})$: negative real part of the least stable eigenvalue fitted from a 2×2 discrete linear map via isotropic salinity perturbations.	Inverse of the long axis of the on-state fluctuation cloud ellipse.
Convergence time	Mean return time of on-state trajectories among 10 000 uniform random initial conditions in a ±2 psu box; convergence at $\varepsilon = 0.05$ psu proximity.	Mean return time of valid on-state trajectories on a 20×20 grid over ±2 psu; convergence at $\varepsilon = 0.05$ psu proximity.	Mean return time of completed trajectories on a 9×9 grid over ±2 psu; convergence defined by permanent entry into an $\varepsilon = 0.05$ psu ball.	Mean transit time from the last edge-state 4σ ellipse visit to the first on-state 4σ ellipse entry.
Basin volume	Fraction of 10 000 random initial conditions in a ±2 psu box centred on the on state that converge to the on state.	Fraction of valid initial conditions on the 20×20 perturbation grid converging to the on state.	Fraction of the 9×9 grid initial conditions over ±2 psu that converge back to the reference equilibrium (permanent entry into an $\varepsilon = 0.05$ psu ball).	Not estimated because the available edge tracks do not thoroughly sample a reference region of state space.
Minimal critical shock	Minimum salinity-plane distance from the on state to a sampled initial condition not converging to the on state.	Minimum salinity-plane distance from the on state to a sampled initial condition not converging to the on state.	Minimum salinity-plane distance from the on-state to a sampled initial condition not converging to the on state (defined only at 315 and 330 ppm).	Gap between the on-state and edge-state 4σ Gaussian ellipses along the line joining their centres in the deep-box salinity plane.

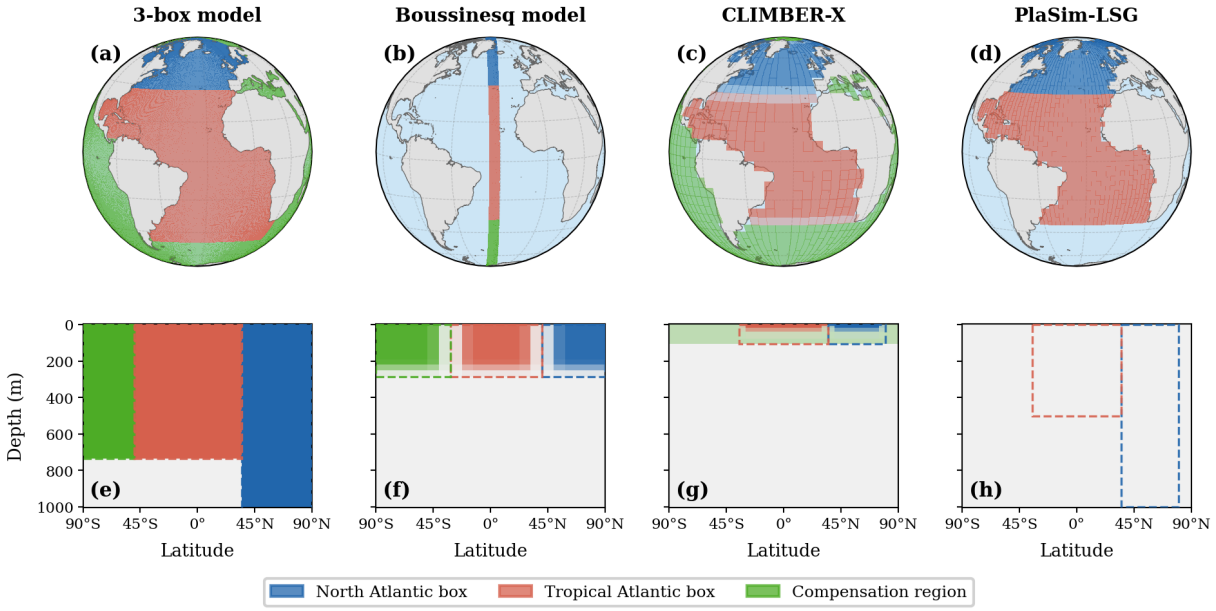


Figure 1: **Perturbation and read-out geometry across the model hierarchy.** This overview compares the geographic salinity regions used as perturbation coordinates (shaded regions) in the Boussinesq and CLIMBER-X experiments and as salinity read-outs (dashed boxes) in the Boussinesq, CLIMBER-X, and PlaSim-LSG analysis. The box geometry associated with the 3-box model is approximately reproduced from the model conception in Wood et al. (2019).

as constants by the parameter vector, while Indo-Pacific salinity is derived at each time from salt conservation. The exact model equations and parameters are given in Appendix A.

Wood et al. (2019) calibrated two distinct parameter sets for the model: one for a pre-industrial climate ($p_{1\times}$) and one for a doubled-CO_2 climate ($p_{1\times}$~~$p_{2\times}$~~) in the FAMOUS Earth System model. We construct a continuum of models through the full parameter vector as $p(r) = p_{1\times} + r(p_{2\times} - p_{1\times})$ and label it by $C(r) = 280\,\text{ppm} + 280\,\text{ppm}\,r$. Thus $r = 0$ and $r = 1$ correspond to 280 and 560 ppm, respectively, while $r > 1$ denotes linear extrapolation beyond the doubled-CO_2 parameter vector. The nominal spacing is $\Delta r = 0.05$, or 14 ppm, and we investigate the system from 280 through 896 ppm, with one additional point at 903 ppm ($r = 2.225$). We find that the on-state vanishes in a bifurcation when extrapolating the linear parameter ramp further, ~~at around 900 ppm~~between 903 and 910 ppm.

All numerical experiments and resilience calculations are performed using the Julia package Attractors.jl (Datseris et al., 2023) within the DynamicalSystems.jl ecosystem.

This computational framework enables efficient integration of large initial-condition ensembles and simultaneous tracking of multiple resilience measures as a parameter changes (Morr et al., 2026). At each continuation step, we probe the two-dimensional state space with 10 000 initial conditions sampled uniformly from a ± 2 psu rectangle centred on the on-state attractor. From these, we derive basin volume as the fraction of trajectories converging to the on state, the minimal critical shock as the minimum distance of the on state to an initial condition outside the basin, and convergence time as the mean time to reach a 0.05 psu neighbourhood. Local resilience is computed from the numerically derived Jacobian of the model's right-hand side at the on-state position. The resilience is measured as $-\mathrm{Re}(\lambda_{\max})$ where $\lambda_{\max}$ is the eigenvalue with the largest real part. In Figure 2, we map out the local basin geometry on a regular grid and show several example trajectories in terms of AMOC strength and phase-space evolution.

2.2 Boussinesq latitude-depth model

The second model in our hierarchy is a latitude-depth Boussinesq model of the AMOC. The model evolves temperature, salinity, streamfunction, and vorticity on a 21×41 grid of discretised latitude and depth points. The one-dimensional sea surface is forced by cosine profiles of temperature and salinity flux, with lower temperatures and lower net freshwater input at high latitudes. This model is much less expensive than global circulation models but already contains spatially structured salinity fields and overturning, making it a useful intermediate step between the conceptual box model and the three-dimensional climate models.

We introduce a single external forcing parameter that mimics Arctic amplification. Increasing it flattens the equator-to-pole surface temperature profile (Arctic amplification) and enhances the freshwater forcing profile (intensification of the hydrological cycle). For the exact mechanism of this parameter, its mapping to CO_2, and the exact model equations, see Appendix B. The perturbation experiments are performed across seven forcing values corresponding to CO_2 concentrations from 280 ppm to approximately ~~424 ppm~~643 ppm. For each configuration, the model is first spun up twice under weak

hemispheric salinity-forcing asymmetries to ensure convergence to the on- and off-states, respectively. Under the symmetric salinity forcing, for which an active and a reversed overturning state can coexist over part of the forcing range, we obtain the final AMOC on and off reference states by evolving the asymmetric-forcing equilibria until convergence. Based on the reference on-state, we initialise perturbed salinity trajectories. We then classify each perturbed run by the reference state it settles to, which in turn enables the basin-volume, minimal critical shock, and convergence-time measures.

Initial-condition perturbations are applied to grid-aligned salinity boxes, and we define an affine conversion of the non-dimensional salinity units of the Boussinesq model to psu (for details see Appendix B). The North Atlantic box covers 40.5°–90°N, the Tropical Atlantic box covers 31.5°S–40.5°N, and the Southern compensation box covers the domain south of 31.5°S. All three extend approximately through the upper third of the water column. The perturbation fields are cosine-tapered to avoid sharp salinity gradients. Perturbations are applied as salinity anomalies $(\Delta S_N, \Delta S_T)$ on a 20×20 grid (400 initial conditions), with both amplitudes spanning ± 2 psu relative to the asymmetric-forcing on-state equilibrium, which lies within 0.1 psu of the final on-state. A compensating perturbation in the Southern box conserves total domain salinity.

For the basin-mapping ensemble, only the salinity field of the on-state equilibrium is carried over into the perturbed initial conditions. The temperature, vorticity, and streamfunction fields are set to zero and left to re-adjust to the perturbed salinity under the model dynamics. This isolates the salinity perturbation as the driver of the response. To estimate local linear stability, we run a second, smaller ensemble of eight experiments per forcing level. Each run has a salinity perturbation in the two-dimensional (North and Tropical Atlantic) salinity space. The eight perturbations are evenly arranged on a circle of radius $\varepsilon = 0.5$ psu around the on-state equilibrium. Unlike the basin-mapping ensemble, these perturbations are applied to the complete equilibrium state, retaining all temperature, vorticity, and streamfunction fields, so that the local recovery of the full system is reflected in the two-dimensional salinity projection. All trajectories are run under neutral symmetric salinity forcing for approximately 500 model years. After

discarding an initial ≈50-year transient, a 2×2 discrete linear map M is fitted by least squares to trajectory pairs $(x_t, x_{t+\Delta t})$ sampled every ≈20 years, where $x = (\delta S_N, \delta S_T)$ denotes the box-salinity deviation from equilibrium. The continuous-time generator $A = \log(M)/\Delta t$ of the fitted map M yields eigenvalues whose least stable real part gives the local resilience $-\mathrm{Re}(\lambda_{\max})$.

Time metrics are reported on a year-proxy scale, set to 100 times the arbitrary model time unit, to facilitate comparability across illustrations. In Figure 3, we plot four exemplary trajectories in terms of the two box salinities and the AMOC strength at 280 ppm and approximately ~~369 ppm~~488 ppm.

2.3 CLIMBER-X

CLIMBER-X (Willeit et al., 2022) is a fast Earth system model of intermediate complexity (EMIC) that couples a 3-D ~~ocean general circulation model, an energy-balance~~ frictional-geostrophic ocean model, a 2.5-D statistical-dynamical atmosphere, a ~~thermodynamic~~ dynamic-thermodynamic sea-ice model, and land-surface components. The ocean component resolves the full three-dimensional salinity and temperature fields, including the main overturning circulation cells, at a resolution of approximately $5° \times 5°$ with 23 vertical levels. CLIMBER-X is computationally efficient enough to permit multi-millennial integrations, making it well-suited for exploring the state space through moderately sized perturbation ensembles.

We first equilibrate CLIMBER-X at 15 prescribed atmospheric CO_2 concentrations between 285 and 600 ppm. Each equilibrium run is integrated for up to 20 000 model years from the default pre-industrial restart. After equilibration, the trailing 500 years of the annual ocean time series are used to store reference values for AMOC strength at 26°N and for the salinity diagnostics in the geographical boxes illustrated in Fig. 1.

The North Atlantic box comprises Atlantic-basin grid cells from 35°–80°N and the Tropical Atlantic box comprises Atlantic-basin cells from 35°S–35°N. Both boxes cover the upper 105 m, corresponding to the top four ocean layers. The North and Tropical box perturbations use a two-grid-cell latitudinal taper and a two-layer vertical cosine taper.

The tapered fields are normalised so that the volume-weighted mean anomaly over the hard target box remains exactly equal to the prescribed salinity perturbation. The salt added to the two Atlantic boxes is compensated uniformly over the rest of the world ocean within the same upper 105 m, so that the global ocean salt content in the restart is conserved.

For our experiments, we apply instantaneous salinity perturbations to these boxes directly to the ocean restart as salinity anomalies in psu, creating an ensemble of modified initial conditions. The North Atlantic and Tropical Atlantic perturbations each take the values $-2, -1.5, \ldots, 2$ psu, giving a 9×9 grid of 81 experiments per CO_2 level. Each perturbed initial condition is integrated forward for 10 000 model years. This model is known to exhibit an AMOC off-state under high CO_2 concentrations when large constant freshwater forcing in the North Atlantic is applied. In the present setup of pure initial salinity perturbations, the off-state is not reached. Instead, at higher CO_2 a second, distinct attractor with a somewhat stronger AMOC emerges. This occurs when sea ice in the Northern Hemisphere retreats and the deep-water convection driving the AMOC expands into areas farther north (Willeit and Ganopolski, 2024). We classify each grid trajectory by its final state using the same criterion as for the ~~Bussinesq~~ Boussinesq model: a run is assigned to the on-state if its final salinities lie within an $\varepsilon = 0.05$ psu ball of the unperturbed reference state, and to the "strong" state otherwise.

A further CO_2-dependence concerns the long equilibration run itself: the state on which it settles, and which serves as the unperturbed reference, changes branch with forcing. Up to 330 ppm it lies on the reference ("modern") branch, whereas from 345 ppm onward it lies on the stronger-AMOC ("strong") branch. The resilience measures obtained from the perturbation experiments are therefore associated with the modern state at 285–330 ppm and with the strong state at 345–600 ppm. The basin volume reported for CLIMBER-X (Table 1) is the share of the 81 grid trajectories that return to this reference equilibrium. It equals one wherever all perturbations recover and falls below one only near the modern-branch transition (at 315 and 330 ppm), where a fraction of perturbations settle on the strong state instead. From the same grid, we obtain the minimal critical

shock as the smallest salinity-plane distance from the reference equilibrium to a perturbed initial condition that does not return to its basin. It is defined only where at least one perturbation escapes, i.e. at 315 and 330 ppm, since at all other CO_2 levels the entire ± 2 psu grid recovers. Figure 4 shows four example trajectories starting from perturbed salinity initial conditions in the specified boxes, similar to the two figures above. For both CO_2 values, the equilibrium states are the modern state.

To estimate local linear stability, we run a second, smaller ensemble of 9 experiments per CO_2 level. Eight runs apply isotropic salinity perturbations of magnitude $\varepsilon = 0.1$ psu again arranged on a circle in the two-dimensional (NA, Tropical) shallow-box salinity space, plus one unperturbed reference run. Each run is integrated for 500 model years. An equilibrium reference is obtained from the mean of the unperturbed run over years 10–150. A 2×2 discrete linear map is fitted by least squares to annual trajectory pairs (x_t, x_{t+1}) in the same fitting window. The continuous-time generator $A = \log(M)/(1\,\mathrm{yr})$ yields eigenvalues whose least stable real part gives the local resilience $-\mathrm{Re}(\lambda_{\max})$.

2.4 PlaSim-LSG

The last component of the hierarchy is the coupled climate model of intermediate complexity PlaSim-LSG, comprising a spectral primitive-equation atmosphere (T21, $\approx 5.6^\circ \times 5.6^\circ$, 10 vertical levels) and the Large Scale Geostrophic (LSG) ocean (Maier-Reimer et al., 1993), a full 3-D ocean circulation model at $3.5^\circ \times 3.5^\circ$ resolution with 22 vertical levels, coupled to a thermodynamic sea-ice scheme and a hydrological cycle. With approximately 10^5 degrees of freedom and a throughput of roughly 700 model years per CPU day, PlaSim-LSG is considerably more complex than CLIMBER-X. In particular, it resolves atmospheric variability more faithfully and thereby yields seemingly chaotic ocean dynamics.

At 285 ppm (pre-industrial) and 360 ppm (contemporary) CO_2, PlaSim-LSG exhibits a bistable AMOC with an on-state of ≈ 15 Sv and an off-state (deep convection in Subpolar Gyre ceases) of $\approx 1 - 2$ Sv, both confirmed by long unforced simulations. Börner et al. (2026) applied an edge-tracking algorithm (Lucarini and Bódai, 2017) to locate the edge

state (Melancholia state) of the AMOC in PlaSim-LSG at 285 ppm and 360 ppm. This is a quasi-stable equilibrium on the complex basin boundary between the on and off states. The algorithm iteratively bisects pairs of trajectories converging to opposite attractors and tracks the resulting pseudotrajectory along the basin boundary until it converges to the edge state. The edge state is a chaotic saddle characterised by persistent centennial AMOC oscillations ($\approx$ 120 yr period, amplitude 3–10 Sv), driven by atmosphere–ice–ocean interactions in the North Atlantic. A complete description of the model configuration, its AMOC bistability, and the edge-tracking procedure is given in Börner et al. (2026), to which we refer the reader for full methodological details.

From the final iterations of edge tracking at each CO_2 level, we extract pairs of edge tracks: one converging to the on-state and one converging to the off-state, both initialised from near-boundary conditions that closely shadow the edge state before diverging. These trajectories provide a natural set of initial conditions from which resilience-relevant transient behaviour can be quantified: they start close to the basin boundary and thus represent challenging recovery paths for the on-state. In total, we analyse 37 paired edge-track files at 285 ppm and 38 at 360 ppm. Dedicated long equilibrium simulations of the AMOC-on, AMOC-off, and edge states at each CO_2 level provide the attractor clouds against which these transients are evaluated.

We reduce the high-dimensional PlaSim-LSG salinity output to the salinity of two geographical boxes. The North Atlantic readout is the mean salinity from 35°–80°N and 0–1000 m depth, and the Tropical Atlantic readout covers 35°S–35°N and 0–500 m depth. Dedicated on-, off-, and edge-state equilibria trajectories determine the means associated with these states in this two-dimensional salinity plane. The covariance matrices of the scattered trajectory points are computed to obtain an approximate Gaussian distribution of data for each state. Convergence time is measured from the last visit to the edge-state 4σ Gaussian ellipse to the first entry into the target-attractor ellipse, and the critical-shock proxy is the gap between the edge-state and on-state ellipses, measured along the line joining their centres. Local resilience is quantified as the inverse width of the on-state fluctuation cloud, $[2000\sqrt{\lambda_{\max}(C)}]^{-1}$, where $\lambda_{\max}(C)$ is the largest eigenvalue of the on-

state equilibrium covariance C. A broader cloud, i.e. larger salinity fluctuations around the on-state, corresponds to lower local resilience since the system appears less capable of absorbing internal variability.

3 Results

The trajectories from the described perturbation experiments are synthesised to give resilience estimates according to all measures listed in Table 1. The evolution of the resilience measures is shown in Figure 6, which forms the basis for the subsequent comparison and differentiation of the results. The models differ substantially in the dynamical characteristics of the AMOC, including its stationarity, multistability, and bifurcations. This, in itself, complicates the interpretation of our results on AMOC resilience within a common framework.

The models do not agree on a consistent impact of atmospheric CO_2 on the equilibrium AMOC strength. The 3-box model exhibits AMOC weakening, which is consistent with behaviour exhibited by the FAMOUS Earth System Model (Bouttes et al., 2015), the calibration target for the 3-box model (Alkhayuon et al., 2019). The vanishing of the AMOC on-state in a bifurcation at a threshold CO_2 concentration has, on the other hand, not been observed in FAMOUS. In the Boussinesq model, CLIMBER-X, and PlaSim-LSG, AMOC strength increases as the CO_2 concentration rises. For the latter two models, it has been conjectured that this is due to a changing net surface freshwater flux, particularly in the Tropical Atlantic (Willeit and Ganopolski, 2024). More evaporation implies more saline water reaching the North Atlantic. Abstracting this mechanism could provide insight into the same phenomenon in the Boussinesq model. Because the surface freshwater forcing weakens in the central (tropical) part of the model relative to the right (northern) part, the ~~sald-advection~~ salt-advection feedback might weaken as a response to the implicit CO_2 forcing. None of the three models exhibits a vanishing AMOC on-state in a bifurcation.

The existence and characteristics of an AMOC off-state are also not consistent across the model hierarchy. The 3-box and Boussinesq models feature stable states with reversed

AMOC flow. That is, the salinity gradient between the North and Tropical Atlantic reverses in both models, and in the Boussinesq model, the actual fluid overturning changes direction. It is unclear whether this is physically meaningful behaviour, i.e. whether any real-world AMOC collapse would lead to a strong reversed flow. However, as we regard alternative stable states primarily as differentiations from the contemporary AMOC state, their exact characteristics are of little concern. The PlaSim-LSG model exhibits an AMOC off-state characterised by a breakdown of deep-water convection in the North Atlantic. The CLIMBER-X model exhibits a weak AMOC state only at CO_2 concentrations lower than preindustrial conditions. This is due to more extensive North Atlantic sea ice blocking deep-water convection in the Labrador and Irminger Seas. The only relevant alternative state for our experiments is a stronger AMOC state at large CO_2 concentrations, reached when diminished Northern sea ice allows for additional deep-water convection in the Kara Sea and Nansen Basin (Willeit and Ganopolski, 2024). It is important to consider these differences, especially when interpreting the results of basin volume and minimal critical shock. The two measures should only be interpreted as resilience against AMOC collapse when the alternative state is indeed associated with an AMOC weakening.

Local resilience of the on-state describes the restoring force that returns small perturbations back to the equilibrium. It can be interpreted as an upper bound on the reciprocal of the half-life of any small perturbation. Critical slowing-down theory predicts a decrease in local resilience as one approaches a bifurcation tipping point. This is indeed what we see for the 3-box model. Local resilience of the Boussinesq and CLIMBER-X AMOC also seems to decrease with increasing CO_2, though the decrease is less pronounced and accompanied by fluctuations in the latter case. In the PlaSim-LSG model, the trajectory cloud forming the quasistationary on-state equilibrium slightly decreases in size, which is consistent with a larger local Lyapunov exponent and, thereby, stronger local resilience.

The convergence time aggregates show consistent trends as CO_2 forcing increases. In all models, the mean time needed for a probed perturbation to converge back to the on-state increases. This can be interpreted as a loss of resilience, as the AMOC is outside its usual operating regime for longer. One model-specific behaviour is worth flagging in

this context. The transient trajectories of some perturbations in the CLIMBER-X model at 315ppm show long occupancy of the strong AMOC state before eventually converging to the modern state. It is possible that all perturbations marked as converging to the strong state in the 10,000 years of trajectory integration would eventually return to the modern state. This highlights the limitations in the numerical computation of resilience measures and the need to accurately define a finite-time framework for their interpretation. Throughout this study, convergence is never determined analytically but obtained via a finite-time cutoff evaluation against a neighbourhood convergence criterion (Morr et al., 2026).

A less intuitive result emerges from the basin volume computations. The share of initial perturbations in the region spanned by ± 2 psu in Northern and Tropical salinity that converge to the on-state rises in the 3-box and Boussinesq model when CO_2 increases. This is surprising insofar as it indicates a strengthening of the AMOC's resilience in this interpretation. Only in the immediate proximity to the bifurcation threshold of the 3-box model do we see a sharp decrease in basin volume. The basin structure of the CLIMBER-X model is almost trivial. For most considered CO_2 concentrations, all probed perturbations converge to either the modern AMOC on-state (300ppm and below) or the emerging strong AMOC state (345ppm onwards). Only at 315 ppm and 330 ppm do perturbations of the modern state lead to a transient convergence to the strong state, splitting the basin.

Finally, the minimal critical shock shows a consistent increase ~~across all models. This~~ in the 3-box, Boussinesq, and PlaSim-LSG models. In CLIMBER-X, it is only defined at 315 and 330 ppm, where it decreases as the strong state takes over a growing part of the probed region. The increase in the other three models would mean an increase in resilience, as larger perturbations are needed to push the AMOC to divergence away from the on-state.

4 Discussion

The picture emerging from our multi-model, multi-measure comparison of AMOC resilience is ambiguous. It is confounded by challenges in conceptual compatibility and

practical differences in dynamical characteristics. Some measures exhibit internal inconsistencies in resilience trends across models. Other measures show a consistent increase in resilience, which seems incompatible with a consistent decrease in resilience on another measure. The notion of a real-world ~~AMOMC~~ AMOC at risk of bifurcation tipping due to anthropogenic influences suggests a decrease in all resilience measures discussed near the bifurcation point (Morr et al., 2026). The simple bifurcating 3-box model does produce a decrease in basin volume and minimal critical shock, but only as it enters the direct proximity of the bifurcation, and the on-state basin contracts to a smaller size than the ± 2 psu salinity region.

The moderate decrease in local resilience observed in the Boussinesq and CLIMBER-X models reinforces the notion that critical slowing down should be expected prior to a bifurcation-mediated AMOC collapse. However, these quantities are obtained in equilibrium, and the real-world AMOC is likely still adapting to fast anthropogenic influences. The direct implications for the applicability of CSD-based early warning signals, therefore, remain limited. In addition, the PlaSim-LSG result suggests a strengthening of local resilience through a reduction in the stationary variance of the quasi-equilibrium state. The consistent increase in non-local convergence time could be interpreted as indicating that the change in the local restoring force generalises to larger regions of the basins of attraction. It lends further credence to the notion of critical slowing down, even as the local radius of perturbations is extended.

Basin volume, as we have computed it here, provides insight into how the on-state provenance evolves under the chosen perturbation regime. For the Boussinesq model, the increase is moderate, as in fact only ~~XXX out of XXX~~ 31 out of 400 perturbation runs switch their basin membership between 280 and 643 ppm, all of them from the off- to the on-state basin. The contraction of the modern state basin in the CLIMBER-X model reveals novel insights into Earth System Model dynamics. To our knowledge, this is the first instance of a model of this size exhibiting bistability with respect to changes in the initial salinity configuration. Typically, modelling studies employ persistent hosing influences to permanently alter surface salinity and facilitate an AMOC collapse.

The possibility that pulse perturbations could similarly lead to the convergence to an alternative stable state, albeit a stronger one, should inspire future research.

The minimal critical shock provides context for the results on basin volume, since the two are mechanistically related. It reveals that the increase in basin volume is at least in part mediated by an expansion of the basin boundary at the point where it is closest to the on-state. This point is a natural candidate for the so-called instanton path to pass through ~~CCC~~[doi:10.1063/1.5084025, doi:10.1017/jfm.2025.248]. This is the most likely path in which a random transition occurs when forced by small noise. It is also related to saddle points or edge states, which are unstable equilibrium points on the boundary ~~CCC~~(Lucarini and Bódai, 2017; Lohmann et al., 2025). The result of the PlaSim-LSG model is consistent with the interpretation of a distancing saddle point on the basin boundary of the 3-box and Boussinesq model. This is because, in this case, we are measuring the distance between the on-state and the edge state, which was directly approximated using an edge-tracking algorithm (Börner et al., 2026). The consistent increase in minimal critical shock should be studied in a broader selection of models going forward, as it reveals important information about the most likely transition path in an AMOC collapse.

Our comparative study of AMOC resilience does not offer concrete physical insight into AMOC dynamics. However, the discrepancies and consistent behaviours found across measures and models inspire a more careful selection of these measures and models. In principle, the evolution of resilience measures could provide valuable insight into the mechanistic changes the AMOC undergoes under climate change. It is not a priori expected that simple models exhibit the same resilience behaviour as complex models, and none may represent the true AMOC dynamics well enough to allow generalisation of our findings in any sense. Our findings offer a first insight into whether and how resilience can be consistently defined and computed across models of varying complexity. Experiments with higher-fidelity models may help resolve some of the discrepancies that arose here. If the real-world AMOC is indeed approaching a bifurcation threshold, we expect insights from the simple 3-box model to be valuable, as many current climate models

struggle to reproduce this threshold ~~CCC~~(Valdes, 2011; McCarthy and Caesar, 2023) [doi:10.1126/sciadv.1601666]. On the other hand, results from simple models should be regarded with caution and treated as strong abstractions.

We hope to inspire future research on the AMOC and other natural systems under the framework of dynamical system resilience. Its system-agnostic nature enables effective comparison of system dynamics, while questions remain about its consistent adaptability across model hierarchies.

A Specifics on the ~~3-Box~~ 3-box salinity model

~~The model tracks North Atlantic~~

State variables and units

The 5-box model of Wood et al. (2019) describes the Atlantic overturning through the exchange of salt between a North Atlantic box (N), a Tropical Atlantic box (T), a Southern Ocean box (S), an Indo-Pacific box (IP), and a deep box (B) below the Atlantic boxes. Temperature is not a state variable. It enters only through the prescribed difference $T_S - T_0$ in the density-driven overturning q. Following Alkhayuon et al. (2019), we retain only the North and Tropical Atlantic ~~salinity anomalies S_N and S_T in the internal~~ salinities as dynamical variables. The Southern Ocean and deep-box salinities vary on much longer time scales and are held fixed, while the Indo-Pacific salinity is diagnosed from conservation of the total salt content C. The resulting system is a two-dimensional, piecewise-smooth ordinary differential equation.

Salinities are written in the anomaly convention $S_i = 100(S_i^{\mathrm{dim}} - S_0)$, where S_i^{dim} is the salinity as a mass fraction and $S_0 = 0.035$. One internal unit therefore corresponds to 10 psu, and the ± 2 psu perturbation region of the main text spans ± 0.2 internal units. In the figures, ~~these~~ internal values are converted ~~into practical salinity~~ as $S_i^{\mathrm{plot}} = 35 + 10 S_i$ psu~~. Southern Ocean salinity~~, and salinity distances are multiplied by 10 to express them in psu. The prescribed S_S and ~~Deep-box salinity S_B are prescribed and~~ S_B are

converted to the same ~~anomaly convention.~~ convention, and the Indo-Pacific salinity is ~~instead diagnosed from total salt content C at every model evaluation:~~ evaluated at every call of the right-hand side as

$$S_{IP} = \frac{100}{V_{IP}}\left[C - \frac{V_N S_N + V_T S_T + V_S S_S + V_B S_B}{100} - S_0(V_N + V_T + V_S + V_{IP} + V_B)\right]. \quad (1)$$

~~The hosing-adjusted freshwater fluxes are~~

Governing equations

The model allows for an additional freshwater hosing H (in Sv), distributed to the North and Tropical Atlantic as $\widetilde{F}_N = F_N + 0.1311 \times 10^6 H$ and $\widetilde{F}_T = F_T + 0.6961 \times 10^6 H$. ~~The continuation used here has~~ We set $H = 0$ ~~in both endpoint parameter vectors. The implemented~~ throughout, so that $\widetilde{F}_N = F_N$ and $\widetilde{F}_T = F_T$. The governing equations are

$$\frac{\mathrm{d}S_N}{\mathrm{d}t} = \begin{cases} \dfrac{Y}{V_N}\left[\dfrac{(q + K_N)(S_T - S_N)}{100} - \widetilde{F}_N S_0\right], & q \geq 0, \\ \dfrac{Y}{V_N}\left[-\dfrac{q(S_B - S_N)}{100} + \dfrac{K_N(S_T - S_N)}{100} - \widetilde{F}_N S_0\right], & q < 0, \end{cases} \quad (2)$$

$$\frac{\mathrm{d}S_T}{\mathrm{d}t} = \begin{cases} \dfrac{Y}{V_T}\Big[\dfrac{q(\gamma S_S + (1-\gamma)S_{IP} - S_T)}{100} + \dfrac{K_S(S_S - S_T)}{100} + \dfrac{K_N(S_N - S_T)}{100} - \widetilde{F}_T S_0\Big], & q \geq 0, \\ \dfrac{Y}{V_T}\Big[-\dfrac{q(S_N - S_T)}{100} + \dfrac{K_S(S_S - S_T)}{100} + \dfrac{K_N(S_N - S_T)}{100} - \widetilde{F}_T S_0\Big], & q < 0, \end{cases} \quad (3)$$

$$q = \frac{\lambda\left[\alpha(T_S - T_0) + \beta(S_N - S_S)/100\right]}{1 + \lambda\alpha\mu}. \quad (4)$$

Here V_i are the box volumes, K_N and K_S are gyre mixing coefficients, F_N and F_T are ~~freshwater fluxes, and Y is the model's time-scaling parameter~~the net freshwater fluxes into the North and Tropical Atlantic, α and β are the thermal and haline expansion coefficients, λ is a hydraulic constant, and μ accounts for the advection of heat by the overturning. The routing parameter γ sets the fraction of the positive overturning return flow ~~passing through~~ that enters the Tropical Atlantic from the Southern Ocean rather

Table 2: Parameters of the 3-box model for the pre-industrial ($p_{1\times}$) and doubled-CO_2 ($p_{2\times}$) calibrations of Wood et al. (2019) and Alkhayuon et al. (2019). Salinities S_S, S_B, and S_0 are given as mass fractions.

Symbol	$p_{1\times}$	$p_{2\times}$	Unit
V_N	0.3261×10^{17}	0.3683×10^{17}	m^3
V_T	0.7777×10^{17}	0.5418×10^{17}	m^3
V_S	0.8897×10^{17}	0.6097×10^{17}	m^3
V_{IP}	2.2020×10^{17}	1.4860×10^{17}	m^3
V_B	8.6490×10^{17}	9.9250×10^{17}	m^3
S_S	0.034427	0.034427	–
S_B	0.034538	0.034538	–
S_0	0.035	0.035	–
F_N	0.384×10^6	0.486×10^6	$\mathrm{m}^3\,\mathrm{s}^{-1}$
F_T	-0.723×10^6	-0.997×10^6	$\mathrm{m}^3\,\mathrm{s}^{-1}$
T_S	4.773	7.919	°C
T_0	2.650	3.870	°C
K_N	5.456×10^6	1.762×10^6	$\mathrm{m}^3\,\mathrm{s}^{-1}$
K_S	5.447×10^6	1.872×10^6	$\mathrm{m}^3\,\mathrm{s}^{-1}$
α	0.12	0.12	$\mathrm{kg\,m}^{-3}\,°\mathrm{C}^{-1}$
β	790	790	$\mathrm{kg\,m}^{-3}$
λ	2.79×10^7	1.62×10^7	$\mathrm{m}^6\,\mathrm{kg}^{-1}\,\mathrm{s}^{-1}$
γ	0.39	0.36	–
μ	5.5×10^{-8}	22×10^{-8}	$°\mathrm{C\,m}^{-3}\,\mathrm{s}$
C	4.4463×10^{16}	4.4735×10^{16}	m^3
Y	3.15×10^9	3.15×10^9	s

than from the Indo-Pacific. The ~~overturning flow~~ time-scaling factor $Y = 3.15 \times 10^9\,\mathrm{s}$ combines the number of seconds per year with the factor 100 of the anomaly convention, so that t is measured in years. The overturning q is ~~reported~~ in $\mathrm{m}^3\,\mathrm{s}^{-1}$ and ~~converted to Sverdrups by division~~ is divided by 10^6 ~~. Its dependence on flow direction~~ to obtain Sverdrups. When q changes sign, the direction of the advective salt transport reverses, which makes the right-hand side discontinuous at $q = 0$.

~~The~~ The two parameter vectors are ~~taken from~~ listed in Table 2.

Continuation in CO_2

Wood et al. (2019) ~~and , who calibrated~~ calibrated the two parameter vectors against FAMOUS simulations at pre-industrial ~~(1×) and doubled-(2×~~and doubled CO_2~~) configurations against the FAMOUS Earth System model. We linearly interpolate and extrapolate the full vectors, including changing~~ . Between the two calibrations, the box volumes, freshwa-

ter fluxes, temperatures, mixing coefficients, ~~hydraulic and routing parameters, and~~ λ, γ, μ, and the total salt content all change, whereas S_S, ~~as described in Section 2.1~~S_B, α, β, S_0, and Y are identical. We move along the straight line $p(r) = p_{1\times} + r(p_{2\times} - p_{1\times})$ in the full parameter space, labelled by $C(r) = 280\,\mathrm{ppm} + 280\,\mathrm{ppm}\,r$. This label is a convenient coordinate rather than a physical calibration, in particular for $r > 1$. The continuation runs from $r = 0$ to $r = 2.4$ in steps of $\Delta r = 0.05$ (14 ppm), with one additional point at $r = 2.225$ (903 ppm). The last regular step with an on-state is $r = 2.2$ (896 ppm). At $r = 2.25$ (910 ppm), only the reversed (off) state is found, so the resilience measures are reported from 280 through 903 ppm.

Attractors are found and continued with the global continuation framework of Attractors.jl (Datseris et al., 2023), using a recurrence-based mapper on a fixed state-space grid with $S_N \in [-1, 1]$ (101 points) and $S_T \in [-0.1, 2]$ (106 points), and a seed-and-match scheme to follow attractors from one parameter value to the next. Trajectories are integrated with the stiff Rosenbrock23 solver at absolute and relative tolerances of 10^{-7}. The on-state is identified at $r = 0$ as the attractor with the largest q and is tracked along the continuation.

Resilience estimation

The non-local measures are not computed on the fixed attractor-search grid. Instead, at every continuation step, we centre a square of half-width 0.2 internal units (± 2 psu) on the current on-state fixed point and draw 10 000 initial conditions uniformly from it. The random seed is fixed for each step to make the results reproducible. Each initial condition is integrated until it comes within $\varepsilon = 0.005$ internal units (0.05 psu) of one of the attractors. If this does not happen within 50 000 model years, the initial condition remains unassigned. This long cut-off is needed near the end of the on-branch, where the on-state recovers very slowly (the e-folding time is about 3000 years at 903 ppm) and convergence can take up to about 18 000 years.

From this ensemble, we obtain the basin volume as the fraction of initial conditions that converge to the on-state, the mean convergence time as the average first time at which

these trajectories enter the ε-neighbourhood of the on-state, and the minimal critical shock as the smallest Euclidean distance between the on-state and a sampled initial condition that does not converge to it. The latter is limited by the sampling density and is therefore an upper bound for the true minimal critical shock. Local resilience is the inverse of the characteristic return time $\tau = -1/\mathrm{Re}(\lambda_{\max})$, where $\lambda_{\max}$ is the eigenvalue with the largest real part of the Jacobian at the on-state fixed point, obtained by forward-mode automatic differentiation. All four measures are computed with the stability-measure routines described in Morr et al. (2026).

For Figure 2, the basins are evaluated on a regular 60×60 grid over the same ± 2 psu window around the on-state, using the same 50 000-year cut-off and a slightly larger proximity threshold of 0.1 psu. The four example trajectories start at offsets of ± 1.6 psu in both coordinates and are shown for the first 3000 model years.

B Specifics on the Boussinesq model

Physical setup

The model ~~represents~~ describes a two-dimensional meridional ~~cross-section of the ocean at~~ section of a single ocean basin, with non-dimensional latitude ~~$x \in [0, 5]$ and depth $z \in [0, 1]$, where $x = 0$ corresponds to~~ $x \in [0, 5]$ and depth $z \in [0, 1]$. Here $x = 0$ is the South Pole, ~~$x = 2.5$ to~~ $x = 2.5$ the equator, and ~~$x = 5$ to~~ $x = 5$ the North Pole~~; $z = 0$~~, while $z = 0$ is the ocean floor and ~~$z = 1$ is the surface~~$z = 1$ the surface. For plotting and for the definition of the boxes, we map x to latitude as $x/5 \cdot 180° - 90°$ and z to a nominal depth of $1000(1 - z)$ m. The zonally averaged flow is ~~spatially two-dimensional and incompressible under~~ incompressible and treated in the Boussinesq approximation, in which density variations are retained only in the buoyancy term. The model goes back to Dijkstra and Molemaker (1997) and was recently studied as a stochastic partial differential equation by Soons, Grafke and Dijkstra (2025) [doi:10.1017/jfm.2025.248] and

Table 3: ~~Default non-dimensional~~ Non-dimensional parameters ~~for~~ and numerical settings of the Boussinesq model.

Symbol	Value	~~Physical meaning~~ Meaning
Pr	1	Prandtl number (ν/κ_T)
Le	1	Lewis number (κ_T/κ_S)
Ra	4×10^4	Rayleigh number
a	5	Domain length (~~pole-to-pole~~pole to pole)
δ	0.05	Surface boundary-layer thickness
τ_T	0.1	Thermal restoring time scale
τ_S	1.0	Salinity flux time scale
β_s	0.1, −0.1, 0	~~Spin-up salinity hemispheric asymmetry~~ Hemispheric salinity-forcing asy
α_s	1	Hydrological intensification coefficient
γ	$0, 0.01, \ldots, 0.06$	Arctic-amplification parameter
M, N	20, 40	Grid intervals (21×41 points)
Δt	10^{-3}	Time step

Esclapez et al. (2026). The velocity field is derived from a streamfunction $\psi(x, z, t)$ via

$$u = \frac{\partial \psi}{\partial z}, \qquad w = -\frac{\partial \psi}{\partial x}, \tag{5}$$

where u is the ~~horizontal (meridional) velocity~~ meridional and w ~~is~~ the vertical velocity~~.~~ ~~The vorticity is $\omega = -\nabla^2\psi$~~, and the vorticity is $\omega = -\nabla^2\psi$. The non-dimensional ~~base parameters and~~ parameters and the numerical resolution are given in Table 3~~; the salinity asymmetry is changed between spin-up and perturbed integrations as described below~~.

Governing equations

Temperature T, salinity S, and vorticity ω evolve according to coupled ~~advection–diffusion~~ advection–diffusion equations with surface forcing:

$$\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} + w\frac{\partial T}{\partial z} = \nabla^2 T + \frac{h(z)}{\tau_T}\big(T_s^\gamma(x) - T\big), \tag{6}$$

$$\frac{\partial S}{\partial t} + u\frac{\partial S}{\partial x} + w\frac{\partial S}{\partial z} = \frac{1}{\mathrm{Le}}\nabla^2 S + \frac{h(z)}{\tau_S}\, S_s^\gamma(x), \tag{7}$$

$$\frac{\partial \omega}{\partial t} + u\frac{\partial \omega}{\partial x} + w\frac{\partial \omega}{\partial z} = \mathrm{Pr}\,\nabla^2\omega + \mathrm{Pr}\cdot\mathrm{Ra}\,\frac{\partial}{\partial x}(T - S), \tag{8}$$

$$\nabla^2\psi = -\omega. \tag{9}$$

~~The surface boundary-layer profile $h(z) = \exp(-(1-z)/\delta)$~~ The profile $h(z) = \exp(-(1-z)/\delta)$ confines the surface forcing to ~~the uppermost ocean. The temperature equation relaxes towards the~~ a thin layer below the surface. Temperature is relaxed towards a prescribed surface profile ~~with~~ on the time scale τ_T~~; a smaller τ_T enforces temperature more strongly towards T_s. For salinity, a fixed flux formulation is used instead of a relaxation, which is physically appropriate since ocean~~. Salinity is instead forced by a fixed flux, reflecting that surface salinity is set by the freshwater exchange with the atmosphere rather than by ~~direct~~ relaxation to a reference ~~profile. A global salinity correction is applied at~~ value. After each time step~~to conserve domain-mean salinity exactly~~, the salinity field is rescaled so that its domain mean remains exactly one. The baroclinic term $\mathrm{Pr}\cdot\mathrm{Ra}\,\partial_x(T-S)$ in the vorticity equation encodes a linear equation of state with density anomalies $\rho \propto -(T-S)$. Horizontal density contrasts thus generate overturning vorticity, and $\mathrm{Pr}\cdot\mathrm{Ra} = 4\times10^4$ controls the strength of this driving relative to viscous dissipation.

The unmodified surface temperature profile is

$$T_s(x) = \tfrac{1}{2}\left[\cos\left(\tfrac{2\pi x}{a} - \pi\right) + 1\right], \tag{10}$$

~~giving $T_s = 0$~~ with $T_s = 0$ at both poles and ~~$T_s = 1$~~ $T_s = 1$ at the equator. ~~This equator-to-pole contrast drives the horizontal density gradient that sustains the overturning.~~ The surface salinity flux is

~~$$S_s(x) = 3.5\cos\left(\tfrac{2\pi x}{a} - \pi\right) - \beta_s \sin\left(\tfrac{\pi x}{a} - \tfrac{\pi}{2}\right),$$~~

$$S_s(x) = 3.5\cos\left(\tfrac{2\pi x}{a} - \pi\right) - \beta_s \sin\left(\tfrac{\pi x}{a} - \tfrac{\pi}{2}\right). \tag{11}$$

~~where the dominant~~ The cosine term represents ~~the atmospheric evaporation–precipitation cycle (~~net evaporation in the tropics ~~,~~ and net precipitation at high latitudes~~) and the~~ . The β_s term ~~introduces a weak hemispheric asymmetry that adds a slight freshwater excess~~ breaks the hemispheric symmetry; for $\beta_s > 0$, it adds freshwater in the northern ~~high latitudes relative to the south.~~

~~The vorticity equation (8) is obtained by taking the curl of the momentum equation. The baroclinic production term $\mathrm{Pr}\cdot\mathrm{Ra}\,\partial_x(T-S)$ encodes the linearised equation of state~~

~~under the Boussinesq approximation: density anomalies $\rho \propto -(T - S)$ create horizontal density contrasts that drive overturning vorticity. The product $\mathrm{Pr} \cdot \mathrm{Ra} = 4 \times 10^4$ governs the balance between advective and diffusive vorticity transport and is the primary control on whether a sustained circulation develops~~and removes it in the southern high latitudes.

Boundary conditions and numerical discretisation

~~No-flux ($\partial_n T = \partial_n S = 0$) and no-slip conditions are imposed on all lateral boundaries. The streamfunction satisfies $\psi = 0$ on all walls, consistent with a closed domain with no net overturning flux through any boundary. The depth~~ All walls are stress-free and impermeable, so $\psi = \omega = 0$ on the boundary, and there is no flux of heat or salt through any wall ($\partial_n T = \partial_n S = 0$). The surface forcing enters only through the interior source terms weighted by $h(z)$. The latitude coordinate is discretised on a ~~non-uniform gridthat concentrates points near the surface and the ocean floor via the mapping~~ uniform grid, and the depth coordinate on a stretched grid

$$z_j = \tfrac{1}{2} + \frac{\tanh\!\big(3(j/N - \tfrac{1}{2})\big)}{2\tanh(3/2)}, \qquad j = 0, \ldots, N, \tag{12}$$

~~and the latitude coordinate uses a uniform grid~~which refines the resolution near the surface and the floor. Spatial derivatives are approximated by second-order finite differences. ~~Equations (6)–(8) are integrated using an implicit-explicit (IMEX) scheme with time step $\Delta t = 10^{-3}$~~Time stepping is semi-implicit with $\Delta t = 10^{-3}$: diffusion and ~~surface-restoring terms~~ thermal relaxation are treated implicitly ~~to avoid the diffusive CFL restriction, while advection terms are treated explicitly. The implicit stepat each time leads to a separable~~ and advection explicitly. Each implicit step, as well as the Poisson problem for ψ, takes the form of a Sylvester equation that is solved with the Bartels–Stewart algorithm. All experiments in this paper are deterministic; the stochastic salinity forcing available in the model is switched off.

The AMOC strength is diagnosed as the mean streamfunction over a small fixed window in the northern basin interior ($x \in \{3.5, 3.75\}$, i.e. 36°–45°N, and $z \approx 0.42$–0.54).

With this convention, the active northern-sinking circulation has positive values and the reversed circulation negative values.

Arctic amplification and CO_2 mapping

To ~~sweep the model across~~ move the model through different climate states, the surface temperature forcing is ~~modified to mimic polar amplification by blending the standard profile~~ blended towards its spatial mean~~:~~ ,

$$T_s^{\gamma}(x) = (1-\gamma)\,T_s(x) + \gamma\overline{T_s}, \tag{13}$$

where $\overline{T_s}$ is the mean of the unmodified ~~temperature~~ profile on the model grid~~and $\gamma \in [0,1]$ measures the degree of Arctic amplification. At $\gamma = 0$~~ . At $\gamma = 0$ the forcing is unchanged~~(pre-industrial);~~ , while increasing γ reduces the equator-to-pole temperature ~~gradient, weakening the baroclinic torque in (8) and thereby reducing~~ contrast and thereby the thermal driving of the overturning ~~.~~

in (8). The same parameter ~~also~~ intensifies the hydrological ~~forcing: under a warming climate the evaporation–precipitation contrast strengthens ("wet-gets-wetter"), which we represent by scaling the symmetric component of the surface~~ cycle. We scale the symmetric part of the salinity flux (11) ~~while leaving the hemispheric~~ and leave the asymmetry unchanged,

$$S_s^{\gamma}(x) = (1+\alpha_s\gamma)\,3.5\cos\!\left(\tfrac{2\pi x}{a} - \pi\right) - \beta_s \sin\!\left(\tfrac{\pi x}{a} - \tfrac{\pi}{2}\right), \tag{14}$$

with ~~hydrological-intensification coefficient $\alpha_s = 1$, so that the~~ $\alpha_s = 1$. The evaporation–precipitation amplitude would ~~double at $\gamma = 1$~~ thus double at $\gamma = 1$ and increases by ~~up to~~ 6% ~~across the sweep ($\gamma \le 0.06$)~~at the largest value $\gamma = 0.06$ used here. Increasing γ ~~thus simultaneously~~ therefore weakens the thermal driving and strengthens the ~~salinity forcing of the overturning~~freshwater forcing at the same time.

~~A physically grounded mapping between~~ We relate γ ~~and~~ to the atmospheric CO_2 concentration C ~~is derived from~~ through the logarithmic forcing ~~formula~~ law and an

estimate of polar ~~ocean-surface amplification:~~

$$\gamma(C) = \frac{(\alpha_{\mathrm{sst}} - 1)\,\mathrm{ECS}}{\Delta T_0} \log_2\!\left(\frac{C}{C_0}\right) = 0.1 \log_2\!\left(\frac{C}{280\,\mathrm{ppm}}\right).$$

sea-surface amplification. If the global-mean surface warming is $\mathrm{ECS} \cdot \log_2(C/C_0)$ and the high-latitude sea surface warms by a factor α_{sst} more, the equator-to-pole temperature difference ΔT_0 is reduced by $(\alpha_{\mathrm{sst}} - 1)\,\mathrm{ECS}\,\log_2(C/C_0)$. Expressing this reduction as a fraction of ΔT_0 gives

$$\gamma(C) = \frac{(\alpha_{\mathrm{sst}} - 1)\,\mathrm{ECS}}{\Delta T_0} \log_2\!\left(\frac{C}{C_0}\right) = 0.05 \log_2\!\left(\frac{C}{280\,\mathrm{ppm}}\right). \tag{15}$$

We ~~adopt~~ use the IPCC AR6 ~~best-estimate equilibrium climate sensitivity $\mathrm{ECS} = 3.0\,^{\circ}\mathrm{C}$~~ best estimate $\mathrm{ECS} = 3.0\,^{\circ}\mathrm{C}$ per CO_2 doubling (Intergovernmental Panel On Climate Change (Ipcc), 2023), a ~~polar ocean-surface amplification factor $\alpha_{\mathrm{sst}} = 2.0$ consistent with observed Arctic SST warming being approximately two- to threefold the global-mean increase, and a~~ pre-industrial Atlantic equator-to-pole ~~SST difference $\Delta T_0 = 30\,^{\circ}\mathrm{C}$ (approximate Levitus climatology value). These choices give $\gamma = 0$ at pre-industrial ($C_0 = 280\,\mathrm{ppm}$) and $\gamma = 0.1$ at doubled ($560\,\mathrm{ppm}$) . The parameter sweep covers seven uniformly spaced values of $\gamma \in [0, 0.06]$ (steps of 0.01)~~ sea-surface temperature difference of $\Delta T_0 = 30\,^{\circ}\mathrm{C}$, and $\alpha_{\mathrm{sst}} = 1.5$. The latter is deliberately smaller than the near-fourfold amplification of Arctic surface air temperature [doi:10.1038/s43247-022-00498-3], since sea-ice insulation and ocean heat uptake strongly damp the warming at the ocean surface. With these choices, $\gamma = 0$ corresponds to $C_0 = 280\,\mathrm{ppm}$ and $\gamma = 0.05$ to $560\,\mathrm{ppm}$. The sweep covers $\gamma = 0, 0.01, \ldots, 0.06$, corresponding to ~~concentrations from $280\,\mathrm{ppm}$ to approximately $424\,\mathrm{ppm}$.~~ 280, 322, 369, 424, 488, 560, and 643 ppm. A sweep at $\gamma = 0.07$ (739 ppm) is not used: there, the model is close to the end of the on-branch, two thirds of the perturbed runs do not settle within the integration limit, and an additional weak cluster of final states appears.

~~Perturbation geometry and physical scaling~~

Reference states and perturbation experiments

~~Each salinity box is a finite set of grid cells whose lateral and lower boundaries are snapped to grid-cell faces~~For each γ, two spin-ups of 20 000 time steps (20 model time units) are run with weak hemispheric asymmetry, $\beta_s = 0.1$ and $\beta_s = -0.1$. Starting from each of them, the model is then integrated under symmetric forcing ($\beta_s = 0$) until it settles. The $\beta_s = 0.1$ spin-up leads to the active (on) state and the $\beta_s = -0.1$ spin-up to the reversed (off) state. All perturbed runs use $\beta_s = 0$.

The perturbation boxes are sets of whole grid cells. The North Atlantic box covers $x \in [3.625, 5]$ (approximately ~~40.5°~~40.5°–~~90°N~~90°N), the Tropical Atlantic box ~~covers~~ $x \in [1.625, 3.625]$ (approximately ~~31.5°S~~31.5°S–~~40.5°N~~40.5°N), and the Southern ~~compensation box covers the~~ box the remaining domain south of $x = 1.625$. All ~~boxes include~~ three boxes contain the cells with $z \geq 1 - 0.28449$, ~~corresponding to~~ which are the top 15 ~~levels and a lower face near~~ grid levels and reach down to about 284 m on the ~~1000 m plotting scale. A cosine taper is applied at~~ nominal depth scale. The perturbation field of each box is tapered with a cosine profile at the interior box edges~~while domain-wall edges~~, while edges at domain walls (the poles and the surface) remain sharp. Each ~~tapered~~ field is normalised so that its mean over the ~~corresponding hard membership mask equals one; adding~~ box equals one. Adding a perturbation of amplitude ΔS ~~therefore~~ thus changes the box-mean salinity by exactly ΔS. ~~North Atlantic~~ The salt added to or removed from the North and Tropical Atlantic ~~perturbations are~~ boxes is compensated in the Southern box~~to preserve the model's domain cell-mean salinity~~, which leaves the domain-mean salinity unchanged.

~~A spin-up with weak hemispheric asymmetry $\beta_s = 0.1$ seeds the symmetric-forcing ($\beta_s = 0$) AMOC-on equilibrium. A second $\beta_s = -0.1$ spin-up similarly seeds the reversed-AMOC equilibrium. At the start of each ensemble, both spin-up states are released under $\beta_s = 0$ dynamics and integrated until they settle. Basin-sweep initial conditions retain only the perturbed~~ The basin-mapping ensemble applies the 20×20 grid of amplitudes

$(\Delta S_N, \Delta S_T) \in [-2, 2]^2$ psu to the salinity field of the $\beta_s = 0.1$ spin-up~~; temperature~~, whose box salinities lie within 0.1 psu of the $\beta_s = 0$ on-state for all γ. Temperature, vorticity, and streamfunction are set to zero ~~. A trajectory is stopped when its box salinities enter the $\varepsilon = 0.05$ psu neighbourhood of either startup equilibrium. Valid~~ in these initial conditions. Each run is integrated under $\beta_s = 0$ for at most 500 000 time steps (500 model time units), and is stopped as soon as its North and Tropical box salinities come within 0.05 psu of either the on- or the off-state. Runs that reach the integration limit or develop negative salinity are excluded; for $\gamma \leq 0.06$, all 400 runs at every forcing level are valid. The final streamfunction values are clustered ~~independently, the strongest cluster is designated AMOC-on, and basin~~ by gap detection (gap threshold 0.2 model units), the cluster with the largest streamfunction is identified as the on-state, and each run is assigned to the nearest cluster. The on-state location in the salinity plane is the median final box salinity of the runs assigned to it.

Basin volume is the fraction ~~assigned to that cluster~~of valid runs assigned to the on-state. Mean convergence time is ~~then evaluated for those AMOC-on trajectories relative to the median final box salinities of~~ the ~~cluster.~~ average, over these runs, of the first time at which the box salinities come within 0.05 psu of the on-state location. Minimal critical shock is the smallest salinity-plane distance between the on-state location and the initial condition of a run that ends in the off-state. Between $\gamma = 0$ and $\gamma = 0.06$, the basin volume rises from 0.71 to 0.79. This change is carried by 31 grid cells that switch from the off- to the on-state basin, while no cell switches in the opposite direction.

Local linear stability

Local resilience is estimated from a separate, smaller ensemble. For each γ, the $\beta_s = 0$ on-state is located by settling from the $\beta_s = 0.1$ spin-up with temperature, vorticity, and streamfunction set to zero. For $\gamma > 0$, the on-state of the previous γ is used as the starting point instead. Eight perturbations of radius 0.5 psu are then applied to the complete on-state, in directions spaced by 45° in the $(\Delta S_N, \Delta S_T)$ plane and again with

Southern-box compensation. In contrast to the basin-mapping ensemble, temperature, vorticity, and streamfunction are kept, so that the trajectories describe the relaxation of the full state. Each trajectory is integrated for 5000 time steps. Directions whose final distance from the equilibrium exceeds the initial radius are discarded. After skipping the first 500 steps, the box-salinity deviations $x = (\delta S_N, \delta S_T)$ are sampled every 200 steps ($\Delta t_{\mathrm{obs}} = 0.2$ model time units), and a 2×2 matrix M is fitted by least squares to all pairs $(x_t, x_{t+\Delta t_{\mathrm{obs}}})$. The generator $A = \log(M)/\Delta t_{\mathrm{obs}}$ gives the characteristic return time $\tau = -1/\max \mathrm{Re}(\lambda(A))$, and the local resilience is $1/\tau$.

Physical scaling

Model salinities are ~~mapped~~ converted to practical salinity ~~units~~ by the affine ~~calibration~~ map $S\,[\mathrm{psu}] = 10.72735\,s + 25.45555$~~,~~. It is fixed by requiring that the North Atlantic and Tropical Atlantic boxes of the pre-industrial ($\gamma = 0$~~, $\beta_s = 0.1$) equilibrium to match~~) $\beta_s = 0.1$ spin-up have salinities of 35.0 and 36.5 psu~~;~~. Perturbation amplitudes convert with the slope alone, so the ± 2 psu ~~perturbation grid therefore corresponds to approximately ± 0.18644 model units . Likewise, the overturning~~ grid corresponds to ± 0.186 model units and the 0.05 psu convergence radius to 0.0047 model units. The streamfunction is converted to Sverdrups by ~~anchoring the~~ mapping the AMOC diagnostic of the same pre-industrial ~~equilibrium value $\psi_0 \approx 1.517$ model units~~spin-up, $\psi \approx 1.517$ model units, to 18 Sv, ~~giving~~ which gives a factor of ~~approximately $11.864\,\mathrm{Sv}/\psi_{\mathrm{model}}$. Convergence and characteristic returntimes are multiplied by 100 to obtain the~~ 11.864 Sv per model unit. The equilibrium AMOC strength shown in Figure 6 is that of the $\beta_s = 0.1$ spin-up; the $\beta_s = 0$ on-state used for the resilience measures is somewhat weaker (about 15 Sv at 280 ppm). Time is reported on a year-proxy ~~scale used in~~ scale of 100 years per model time unit. On this scale, the spin-ups last 2000 years, the basin-mapping runs are limited to 50 000 years, and the ~~synthesis, and local resilience is divided by 100 to obtain inverse year-proxy units.~~ local-stability trajectories last 500 years with observations every 20 years. Local resilience is correspondingly divided by 100.

C Specifics on the CLIMBER-X model

Model

CLIMBER-X v1.0 (Willeit et al., 2022) couples the 2.5-dimensional semi-empirical statistical-dynamical atmosphere model SESAM, the three-dimensional frictional-geostrophic ocean model GOLDSTEIN, the dynamic-thermodynamic sea-ice model SISIM, and the land surface model PALADYN. All components run on a regular $5° \times 5°$ grid, and the ocean has 23 vertical layers of increasing thickness, with a top layer of 10 m. On a single node of a high-performance computer, the model simulates about 10 000 years per day. This makes ensembles of multi-millennial integrations feasible, which are out of reach for general circulation models. Atmospheric CO_2 is prescribed as a constant in all our runs.

Equilibria and diagnostics

For each of the 15 CO_2 levels (285, 300, 315, 330, 345, 360, 385, 400, 430, 460, 490, 510, 540, 570, and 600 ppm), the model is started from the default pre-industrial restart and integrated for 20 000 years at fixed CO_2. The final restart serves as the reference state for all perturbation experiments at that level, and the trailing 500 years provide the reference AMOC strength. We diagnose the AMOC as the maximum of the Atlantic overturning streamfunction at 26°N.

The salinity diagnostics are written as annual means by a small modification of the model output routines. The North Atlantic box consists of all Atlantic-basin cells between 35° and 80°N, and the Tropical Atlantic box of all Atlantic-basin cells between 35°S and 35°N. Both boxes are defined by the model's basin mask, so their zonal extent follows the coastlines and is not bounded by fixed longitudes. Both include the top four ocean layers (layer centres at 5.3, 18.4, 39.4, and 78.8 m), which end at about 105 m. The box salinities are volume-weighted means and are also written for the initial state, so the time series start with the applied perturbation.

Perturbations

Perturbations are applied directly to the salinity field of the restart file. Within each box, the anomaly is tapered with a cosine profile over two grid cells at the northern and southern box edges and over the lower two of the four layers. No tapering is applied in longitude, where the box is bounded by land. The tapered field is rescaled so that the volume-weighted mean anomaly over the box equals the prescribed value exactly. To conserve the global salt content, a uniform compensating anomaly

$$\Delta S_c = -\frac{\Delta S_{NA} V_{NA} + \Delta S_T V_T}{V_{\mathrm{glob}} - V_{NA} - V_T} \tag{16}$$

is applied to the rest of the world ocean in the same upper 105 m, where V_{glob} is the volume of the upper 105 m of the global ocean and V_{NA}, V_T are the box volumes.

In the main experiment, ΔS_{NA} and ΔS_T each take the values $-2, -1.5, \ldots, 2$ psu, which gives 81 runs per CO_2 level and 1215 runs in total. Each run is integrated for 10 000 years at the CO_2 level of its reference state.

Classification and resilience measures

The resilience measures are evaluated in the plane of the two box salinities, relative to the final state of the unperturbed grid run. A run is considered converged if it ends within $\varepsilon = 0.05$ psu of this reference state. Its convergence time is the time of the last entry into this ε-ball, after which the trajectory remains inside. Runs that do not end inside the ball are counted as not returning. They are excluded from the mean convergence time and enter the basin volume and minimal critical shock as members of a different basin. The minimal critical shock is the smallest initial distance from the reference state among these runs. Because it is resolved only to the 0.5 psu grid spacing, it is an upper bound on the true value.

At 285, 300, and all levels from 345 ppm upward, all 81 runs return. At 315 ppm, 21 runs settle on the strong state (basin volume 0.741, minimal critical shock 0.713 psu), and at 330 ppm, 46 runs do (basin volume 0.432, minimal critical shock 0.499 psu). The final states are tightly clustered. Within each cluster, they scatter by about 0.002 psu, while

the strong state lies 0.068 psu (315 ppm) and 0.076 psu (330 ppm) from the reference state in North Atlantic salinity. No run ends between 0.02 and 0.05 psu from the reference state, so the classification does not depend on the exact choice of ε between about 0.01 and 0.05 psu. A threshold of 0.1 psu or more, in contrast, would merge the two states. The strong-state AMOC is 0.21 Sv (315 ppm) and 0.36 Sv (330 ppm) stronger than the reference.

The strong state reached from the perturbed runs at 330 ppm has the same North Atlantic salinity (34.587 psu) as the unperturbed equilibrium at 345 ppm (34.585 psu). From 345 ppm onward, the 20 000-year equilibration itself ends on this stronger branch. The resilience measures therefore refer to the modern state at 285–330 ppm and to the strong state at 345–600 ppm. In the latter range, no perturbation within the ± 2 psu grid leaves the strong-state basin, so the minimal critical shock is not defined there.

Local linear stability

For local resilience, nine further runs are performed at every CO_2 level, 135 in total. One run is unperturbed, and eight apply perturbations of 0.1 psu in directions $\theta = 0°, 45°, \ldots, 315°$, i.e. $(\Delta S_{NA}, \Delta S_T) = 0.1\,\mathrm{psu}\,(\cos\theta, \sin\theta)$, with the same global compensation as above. Each run is integrated for 500 years. The equilibrium is taken as the mean box salinity of the unperturbed run over years 10–150, and the deviations x_t of the eight perturbed runs from it are used in the same window. A 2×2 matrix M is fitted by least squares to all annual pairs (x_t, x_{t+1}), and the generator $A = \log(M)/(1\,\mathrm{yr})$ yields the characteristic return time $\tau = -1/\max \mathrm{Re}(\lambda(A))$ and the local resilience $1/\tau$.

D Specifics on the PlaSim-LSG model

Model and simulations

PlaSim-LSG couples the atmospheric general circulation model of the Planet Simulator [doi:10.1127/0941-2948/2005/0043] to the Large Scale Geostrophic ocean model (Maier-Reimer et al., 1993). The atmosphere is a spectral primitive-equation model at T21

resolution (about 5.6° × 5.6°) with 10 vertical levels. The ocean has an effective horizontal resolution of 3.5° × 3.5° and 22 vertical layers, whose thickness increases from 50 m at the surface to 1000 m in the deep ocean. Sea ice is represented by a thermodynamic zero-layer model without ice transport. With about 10^5 degrees of freedom, the model runs at roughly 700 model years per day on a single CPU (Börner et al., 2026). Its AMOC exhibits pronounced multicentennial variability (Mehling et al., 2023).

We use the simulations of Börner et al. (2026) at 285 and 360 ppm, available at doi:10.5281/zenodo.17053348. At both levels, the AMOC is bistable. The on-state has a mean AMOC of 14.7 Sv (285 ppm) and 15.0 Sv (360 ppm), and the off-state, in which deep convection in the subpolar North Atlantic ceases, has 1.1 and 1.7 Sv. Here, the AMOC strength is the maximum of the Atlantic overturning streamfunction between 46° and 66°N, averaged over the full equilibrium runs. The off-state at 360 ppm varies on multicentennial time scales, with 200-year means between 1.4 and 2.2 Sv.

The edge state was located with an edge-tracking algorithm (Lucarini and Bódai, 2017). Starting from a pair of states that converge to opposite attractors, nine intermediate initial conditions are generated by linear interpolation and integrated in parallel. The two neighbouring members that end up on different sides of the basin boundary define the next, ten times closer pair. Once a pair diverges by more than a set threshold in 10-year smoothed AMOC strength, it is refined again in the same way. Concatenating the segments yields a pseudotrajectory that follows the basin boundary towards the edge state, and each iteration also produces two trajectories that leave the boundary towards the on- and the off-state, respectively. These are the edge tracks analysed here. In addition, dedicated long simulations of the on-state, the off-state, and the edge state are available at both CO_2 levels.

Salinity readouts

We reduce the salinity output to two volume-weighted box means. The North Atlantic readout averages over 35°–80°N and the upper 13 ocean layers (down to 1025 m), and the Tropical Atlantic readout over 35°S–35°N and the upper 8 layers (down to 500 m). The

edge-track output is already zonally averaged over the Atlantic. For the three-dimensional equilibrium output, we first apply the Atlantic basin mask of the model (with a southern border at 34°S) and take the zonal mean. Cells are weighted by cos(latitude) times layer thickness, and land cells are excluded.

At 285 ppm, we use 37 edge-track files and, at 360 ppm, 38 files, each containing one track towards each attractor. We classify the tracks by their final state rather than by their label in the edge-tracking output. The North Atlantic salinity averaged over the last 10% of each track is compared with its median over all tracks at that CO_2 level, and tracks above the median are assigned to the on-state. This disagrees with the original labels for two tracks per CO_2 level, none of which converges to its attractor according to the criterion below. From the edge-state equilibrium runs, the first 4 years (285 ppm) and 60 years (360 ppm) are discarded.

Resilience measures

For each of the three states, we compute the mean μ and covariance matrix C of the corresponding equilibrium run in the plane of the two salinity readouts. A point x lies within the 4σ ellipse of a state if its Mahalanobis distance $\sqrt{(x-\mu)^\top C^{-1}(x-\mu)}$ is at most 4. The convergence time of a track is the time between its last visit to the edge-state ellipse and its first subsequent entry into the ellipse of its target attractor. Tracks that never enter the edge ellipse, or never reach the target ellipse afterwards, are counted as not converged. Many tracks at 285 ppm are short, and only 4 of 37 tracks converge to each of the on- and off-states. At 360 ppm, 19 of 38 tracks converge to the on-state and 33 of 38 to the off-state.

The distance between the edge state and an attractor is the gap between their 4σ ellipses along the line connecting the two centres, $\max(0, |\mu_{\mathrm{att}} - \mu_{\mathrm{edge}}| - r_{\mathrm{edge}} - r_{\mathrm{att}})$, where r is the 4σ radius of each ellipse in that direction. For the on-state, this is the minimal-critical-shock proxy of the main text. The local-resilience proxy is $[2000\sqrt{\lambda_{\max}(C)}]^{-1}$, where $\lambda_{\max}(C)$ is the largest eigenvalue of the on-state covariance. The quantity $2\sqrt{\lambda_{\max}(C)}$ is the long axis of the 1σ ellipse in psu, and the additional factor 1000 expresses it in

Table 4: Resilience quantities for PlaSim-LSG in the deep-box salinity plane.

CO_2	State	Mean conv. time (yr)	Edge distance (psu)	Local resilience	AMOC (Sv)
285 ppm	on	105.3	0.050	0.0357	14.7
285 ppm	off	116.0	0.204	0.0115	1.1
360 ppm	on	150.6	0.110	0.0368	15.0
360 ppm	off	74.8	0.093	0.0087	1.7

units of 10^{-3} psu. This proxy reflects the size of the fluctuations around the on-state and is not an eigenvalue of the underlying dynamics. Basin volume is not estimated, since the edge tracks start close to the basin boundary and do not sample a reference region of state space. Table 4 lists all resulting values.

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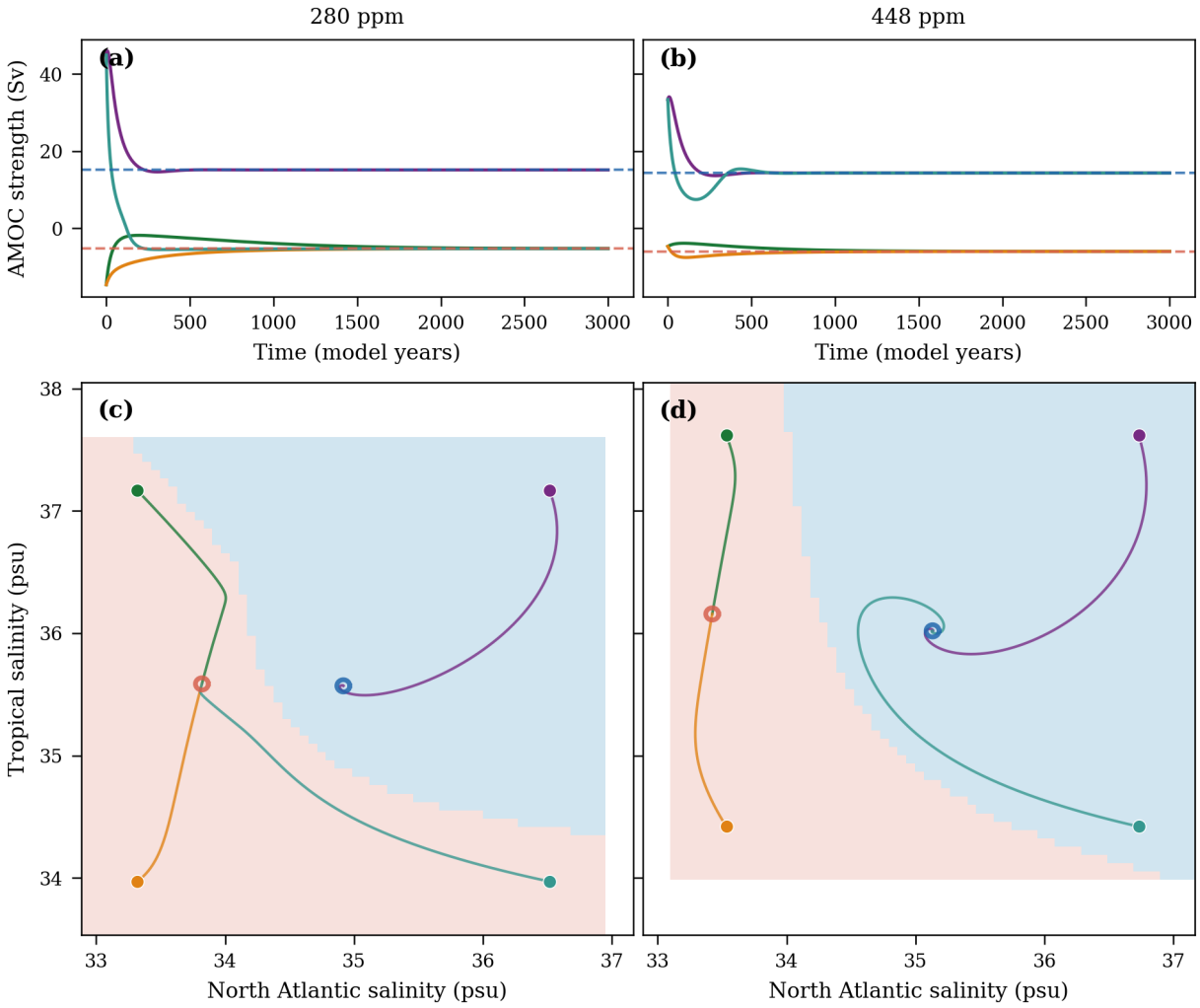


Figure 2: **Three-box model response to salinity perturbations.** Top panels (a, b) show the evolution of the AMOC strength for four representative initial conditions, with dashed horizontal lines marking the on- and off-state equilibria. Bottom panels (c, d) show the same trajectories in the two-dimensional salinity state space (the two state variables), overlaid on the sampled basins of attraction; circles represent 0.05 psu neighbourhoods around the on- and off-state fixed points (blue and orange, respectively). The left and right columns correspond to 280 and 448 ppm model specifications, respectively.

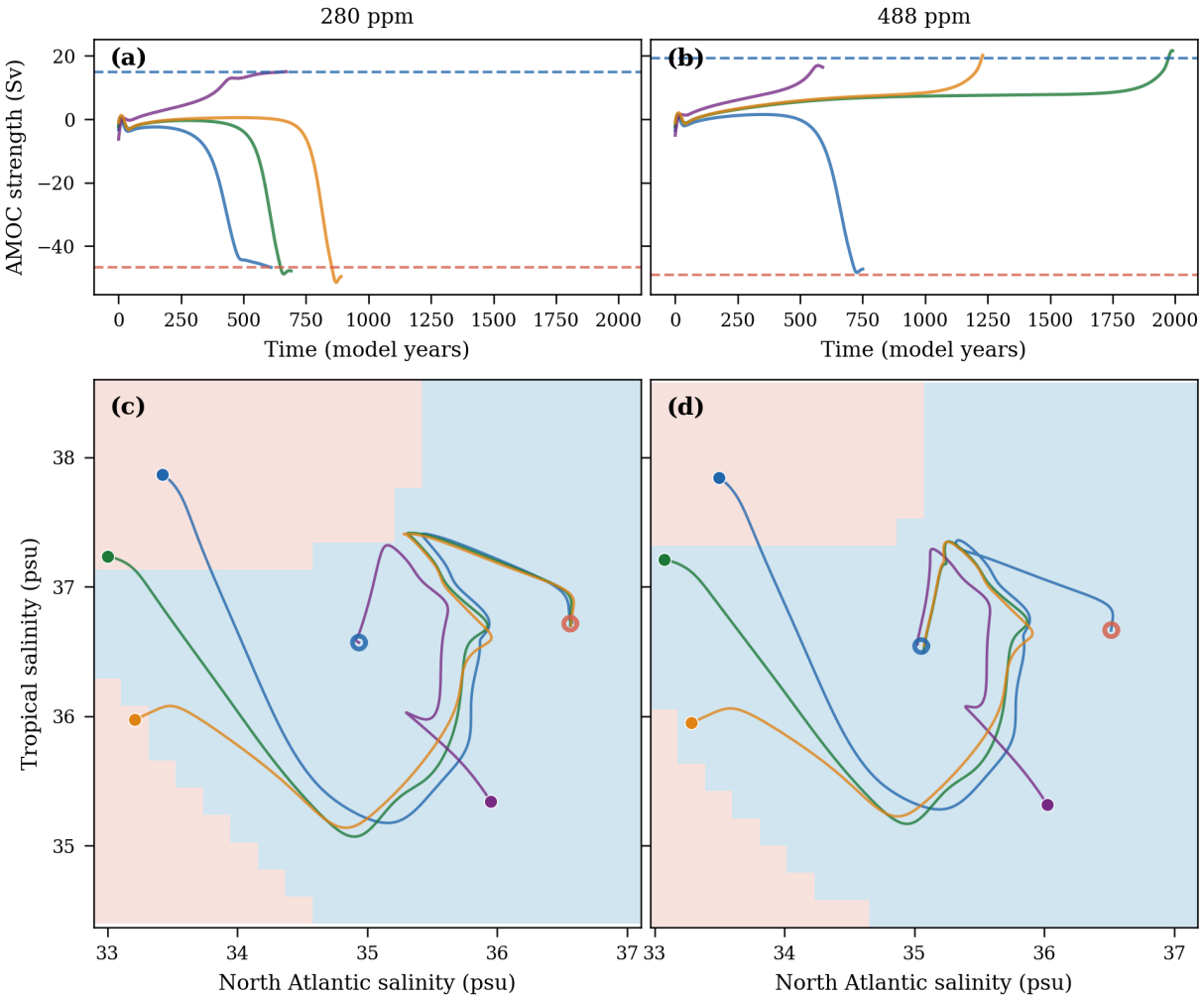


Figure 3: **Boussinesq model response to salinity perturbations.** Top panels (a, b) show the evolution of the AMOC strength for four representative initial conditions (the first 50 years of initial adjustment are omitted), with dashed horizontal lines marking the on- and off-state equilibria. Bottom panels (c, d) show the same trajectories in North Atlantic and Tropical Atlantic box-mean salinity, overlaid on the sampled basins of attraction. Circles represent 0.05 psu neighbourhoods around the on- and off-state fixed points (blue and orange, respectively). The left and right columns correspond to 280 and ~~369 ppm~~ 488 ppm model specifications, respectively.

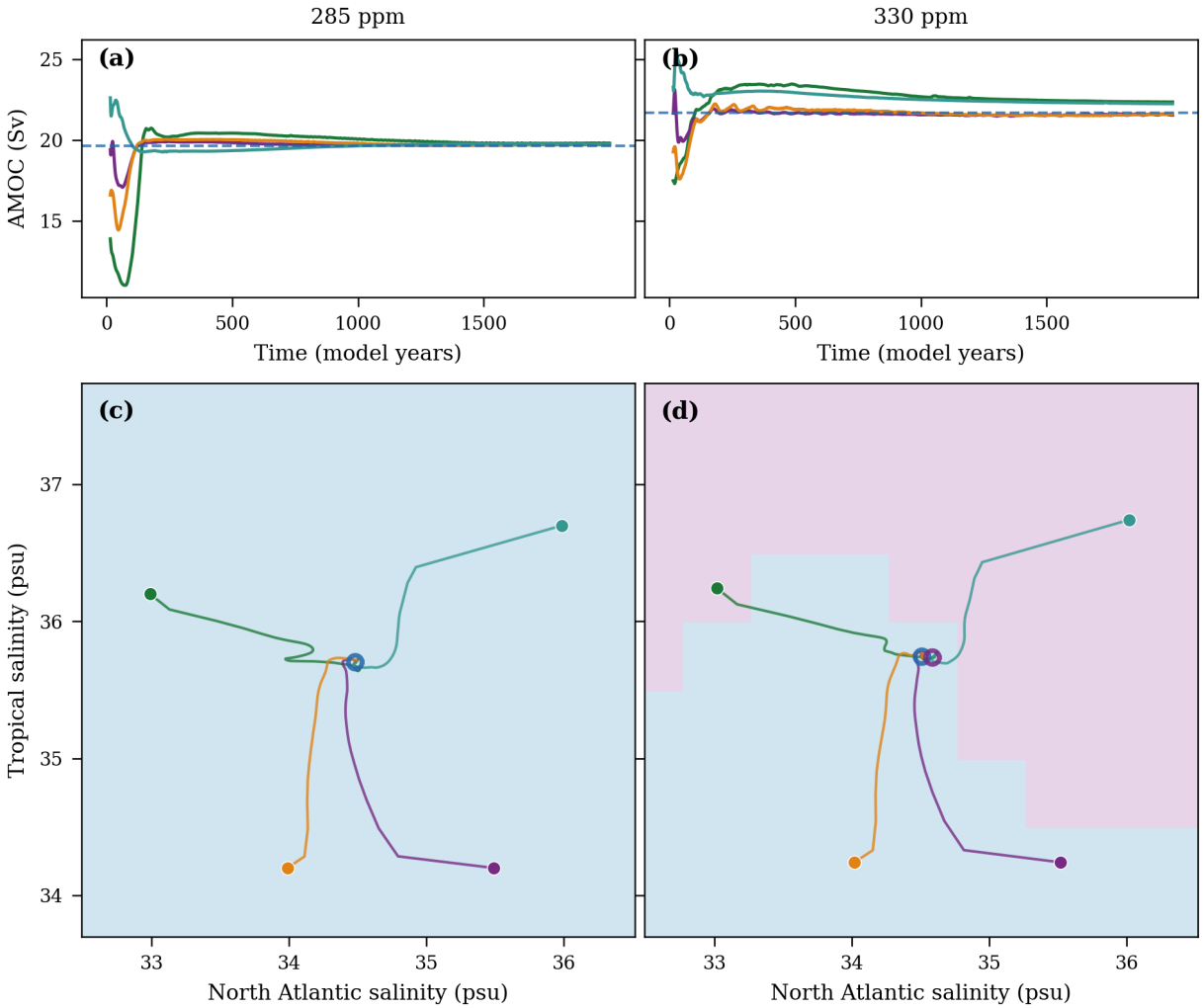


Figure 4: **CLIMBER-X model response to salinity perturbations.** Top panels (a, b) show the AMOC strength at 26°N along four example trajectories from perturbed salinity initial conditions. Bottom panels (c, d) show the same trajectories in the two-dimensional Atlantic-box salinity plane, with axes spanning exactly the ±2 psu perturbation window. The shaded background maps the basin of attraction of every initial condition on the full 9×9 grid: cells whose trajectory returns to the reference on-state (final salinity within $\varepsilon = 0.05$ psu) are blue, and cells that settle on the distinct stronger-AMOC ("strong") state are purple. Circles mark the $\varepsilon = 0.05$ psu neighbourhoods of the on-state (blue, at the unperturbed equilibrium) and of the strong state (purple, at the mean endpoint of the non-converged trajectories). The two columns show prescribed atmospheric CO_2 concentrations of 285 ppm and 330 ppm: at 285 ppm all 81 initial conditions return to the on-state, whereas at 330 ppm ~~45~~ 46 of the 81 converge to the strong state, which lies only ~~≈0.05–0.08 psu~~ ≈0.08 psu from the on-state but exhibits an AMOC about 0.4 Sv stronger. ~~TODO change figure script so that~~ Dashed horizontal lines in (a~~purple dashed line shows~~, b) mark the ~~strong~~ equilibrium AMOC strength of the on-state (blue) and, in ~~panel~~ (b), of the strong state (purple).

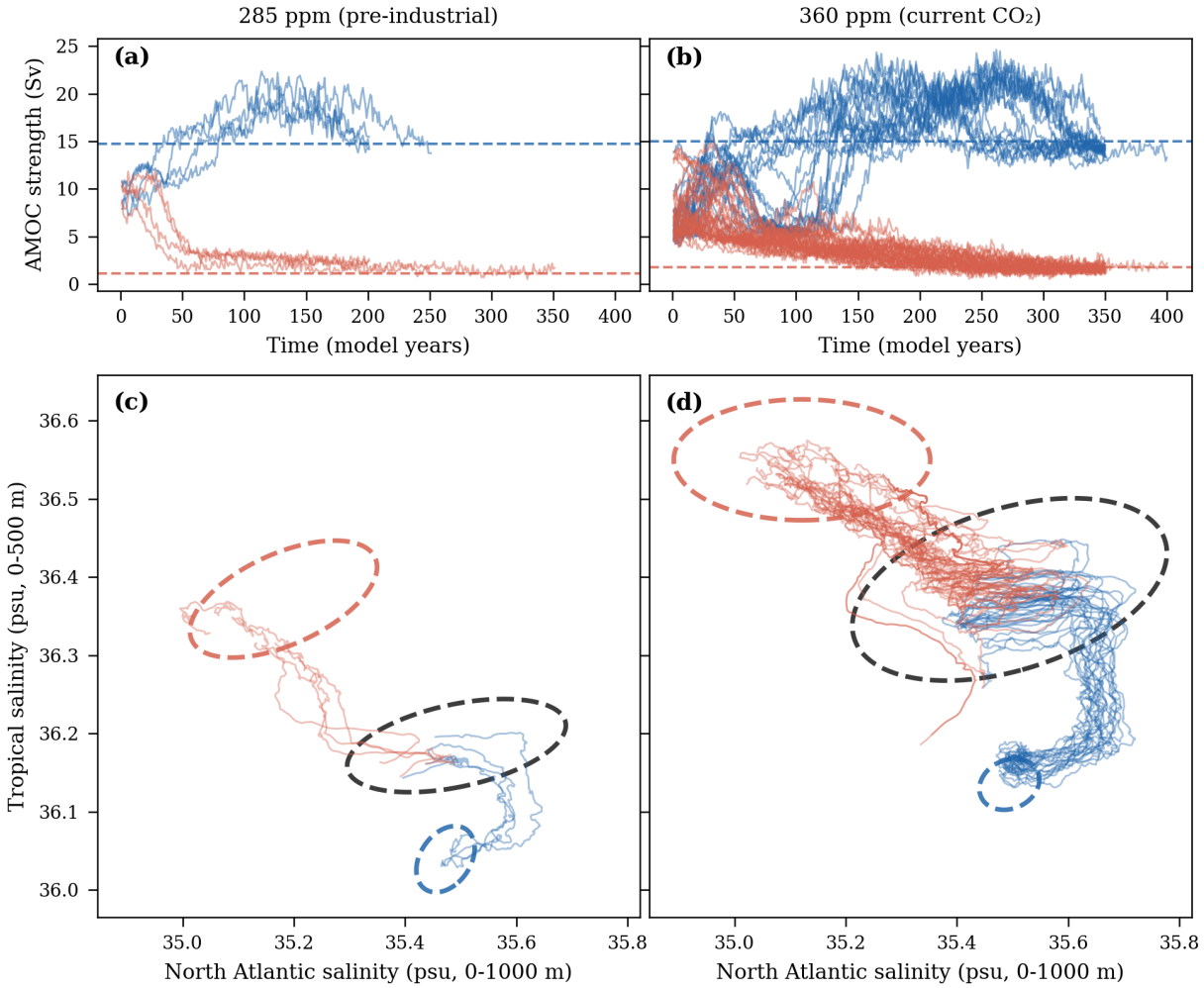


Figure 5: **PlaSim-LSG edge-state trajectories and covariance ellipses of attractors.** Top panels (a, b) show the AMOC strength readout along converged edge-track trajectories. Bottom panels (c, d) show the same transients in the North and Tropical Atlantic salinity coordinates. Ellipses denote the fitted 4σ Gaussian regions to equilibrium runs of the three states (on: blue, off: orange, edge: black), used to define convergence times and edge-to-attractor distances. Trajectories are coloured by the state they converge to. The two columns show the experiment results at prescribed atmospheric CO_2 concentrations of 285 ppm and 360 ppm, respectively. ~~TODO change the figure script so that the color of the off state ellipses, trajectories etc is the same as in the box and boussinesq model.~~

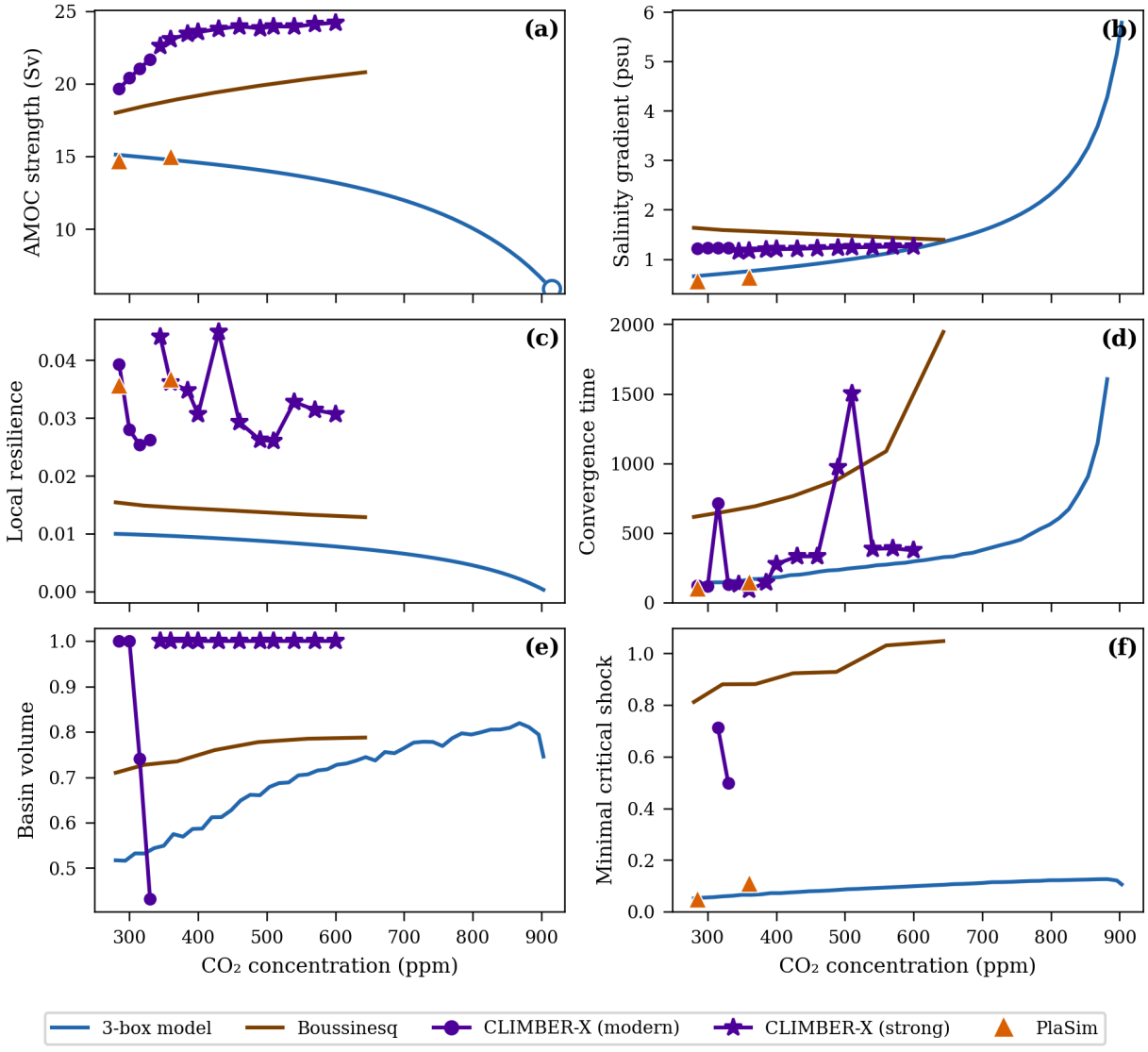


Figure 6: **Cross-model synthesis of AMOC resilience measures.** Equilibrium AMOC strength (a) and Tropical minus North Atlantic salinity gradient (b). Resilience quantifiers are plotted against the CO_2 coordinate for each model, where available (c–f). The model-specific definitions and implementation of each resilience quantifier are given in Table 1. These results provide the basis for assessing which resilience signals are coherent across the model hierarchy.